

Практикум A1

135 marks of the archive, nine techniques, thirteen tasks. Everything the archive asks about arithmetic sequences — the terms, the sums, and the three Paper 3 investigations built on top of them.

О практикуме

Тема `number_algebra.sequences` взята не целиком: в ней лежат обе прогрессии сразу.

Разрез между A1 и A2 идёт по вопросу, а не по теме — у майской 2023 TZ1 бумаги пункты (a)–(c) арифметические, (d)–(e) геометрические, и это один вопрос 10. Доля 0.8 посчитана по баллам: 135 из 169.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-a1-arithmetic-sequences

Собрано из ноутбука репозитория `ib-math-aa-hl-past-papers`. Проверки в PDF не работают — они для интерактивной версии.

A1 — Arithmetic sequences and sums

135 marks of the archive, nine techniques, thirteen tasks. Everything the archive asks about arithmetic sequences — the terms, the sums, and the three Paper 3 investigations built on top of them.

The one idea

An arithmetic sequence is not a formula. It is one sentence:

From each term to the next you add the same thing.

Everything in the topic comes out of that sentence in one line, and coming out is quicker than remembering.

$$u_n = u_1 + (n - 1)d$$

reads *"you have added d exactly $n - 1$ times"*. The -1 is not a quirk of the notation; it is the count of steps. Between the first term and the seventh there are

six steps, not seven, and that off-by-one is the single most expensive slip in the topic.

$$S_n = \frac{n}{2} (u_1 + u_n)$$

is the story about Gauss. Write the sum forwards, write it again backwards underneath, and add column by column: every column gives $u_1 + u_n$, and there are n of them. So twice the sum is $n(u_1 + u_n)$. Substituting $u_n = u_1 + (n - 1)d$ turns it into the other printed form, $S_n = \frac{n}{2} (2u_1 + (n - 1)d)$ — the same formula, not a second one.

The other idea

After that, the topic stops being about formulas at all.

A progression is two numbers, u_1 and d . Almost every question gives you two facts about them.

"The 7th term is 6, and the 6th and 12th add to 24." Two facts, two unknowns, two linear equations — and solving those you

have been able to do since school. The work is writing them down, not solving them.

How the checks work

They do not know the answers, and they do not know the formulas. Each check is handed **the rule** — a first term and a step — and walks:

```
verify_term('4', 8, progression(q4u, q4d), 8)
```

is *"your first term and your common difference: is the eighth term of that sequence equal to 8?"* The term is reached by adding d seven times. $u_1 + (n - 1)d$ appears nowhere, so any correct form of an answer passes: $2n + 3$, $5 + 2(n - 1)$ and $n + (n + 3)$ are the same answer.

The checks run in both directions, and you will use both.

direction	what is known	what carries your answer
forward	the progression	the term, or the sum
backward	the term, or the sum	the progression

The line above is backward: nothing on it is your answer except what is inside `progression(...)`, and the 8 came from the question. That is how *find the first term and the common difference* gets checked without a single answer being stored.

A consequence worth having: **progressions may be built from your own earlier answers.** In task 6 the sums come from your p and q ; in task 3 the rows come from your rule for one row. A wrong first part therefore costs you the first part only — which is what a markscheme calls follow through.

When you are wrong the check says **how**:

what you wrote	what the check says
$u_1 + nd$	the step is taken n times instead of $n - 1$
the term next door	the index is out by one
u_n where S_n was asked	that is the term, not the sum
$n(u_1 + u_n)$	the sum is doubled: the $n/2$ has become n
$u_1 - u_2$	the sign is the other way round
$n = 0$	an index counts terms: it has to be at least one

Order of work

level	what it means	tasks
	the progression is given; walk it	1-3
	the progression is what you are looking for	4-10
	n itself is the unknown, and it is not free	11-13

Every task is a real past-paper question, cited.

47% of these marks are on a calculator paper, and the number lies

in both directions. Twenty-five of those marks are one Paper 3 investigation that is pure algebra — the paper simply allows a calculator. The other way round, task 12 cannot be done without one: "the smallest number that is both triangular and pentagonal" is not an equation, and every markscheme for it says *uses a table of values*.

```

import sys
sys.path.append('.')          # from
practicum/number_algebra to practicum/kit.py
import sympy as sp           # the escape
hatch: anything not in kit is in sp
from kit import *             # checks +
progression, term, total, ln, pi, ...

language('en')               # this
notebook is in English, and so are the checks

n = symbols('n')              # the index:
every one of these papers calls it n
d, p, q, m = symbols('d p q m') # letters
that stay letters
x = symbols('x', positive=True) # every
paper here says  $x > 1$ , and saying so
                                # is what lets
 $\ln(1/x^{*3})$  become  $-3*\ln(x)$ 
                                # k already
comes from kit

# Every progression below may be built from
an answer you have not written
# yet. The two helpers return an empty
progression in that case, so the
# notebook runs top to bottom while it is
still blank and the checks print
# a white square.

def terms(rule):
    # the progression whose nth term is
    rule(n)
    try:
        first, second = rule(1), rule(2)

```

```

        except TypeError:                                     #
an answer inside the rule is blank
            return progression(..., ...)
        if blank(first, second):
            return progression(..., ...)
        return progression(first, second - first)

def sums(rule):
    # the progression whose first n terms add
up to rule(n)
    try:
        first, second = rule(1), rule(2)
    except TypeError:
        return progression(..., ...)
    if blank(first, second):
        return progression(..., ...)
    # the first term is S1, and the second is
S2 - S1, so the step is
    # S2 - 2*S1: nothing here knows the
formula for a sum
    return progression(first, second - 2 *
first)

print('ready; sympy', sp.__version__)
print('a progression:', progression(3, 2))
print('its 5th term: ', term(progression(3,
2), 5))
print('and S5:      ', total(progression(3,
2), 5))

```

Map of the nine techniques

#	technique	you recognise it by	it reduces to
1	which term is it	u_1 and d known, a term or an index asked	$u_n = u_1 + (n - 1)d$
2	add the first n	the word <i>sum</i> , a Σ , something piling up	$S_n = \frac{n}{2}(u_1 + u_n)$
3	two facts, two unknowns	neither u_1 nor d is given	two linear equations
4	from S_n back to u_n	the question says S , the answer wants u	$u_n = S_n - S_{n-1}$
5	the constant difference as a condition	a letter inside the terms, or <i>show that ... is arithmetic</i>	$u_2 - u_1 = u_3 - u_2$
6	quantities with no index	a coefficient, a root and a constant put in sequence	the same condition, on letters
7	the greatest sum	<i>maximum</i> beside S_n , and $d < 0$	the terms turn negative
8	n has to be a whole number	<i>smallest, no cards left over</i>	a table, not an equation

#	technique	you recognise it by	it reduces to
9	terms that are logarithms	$\ln x + p \ln x + \dots$	log laws first, then technique 2

Techniques 1-2 are the two formulas. Technique 3 is the topic's real question and technique 4 is it read backwards. Technique 5 turns the progression from a list into a condition, and technique 6 is the same condition applied to things that have no index at all. The last three are what the exam does when the algebra stops: 7 needs the sign of the terms, 8 needs a table, 9 needs the log laws before anything else.

Part 1. Walking the progression

Theory: two formulas, and where they come from

The term. Adding d once takes you from u_1 to u_2 ; adding it $n - 1$ times takes you to u_n :

$$u_n = u_1 + (n - 1)d$$

When the paper hands you the sequence already written as a formula in n — " $u_n = 15 - 3n$ " — read it the same way. It is linear in n , the coefficient of n **is** the common difference, and the constant term is u_0 , which is not a term of the sequence at all.

The mark that goes missing here: for $u_n = 15 - 3n$ the first term is 12, not 15.

The sum. Write S_n forwards and backwards and add:

$$\begin{array}{rcl}
 S_n & & = u_1 \\
 & + & (u_1 + d) \\
 & + & \dots \\
 & + & u_n \\
 S_n & & = u_n \\
 & + & (u_n - d) \\
 & + & \dots \\
 & + & u_1
 \end{array}$$

Every column adds to $u_1 + u_n$, and there are n columns, so

$$\begin{aligned}
 2S_n &= n(u_1 + u_n) \\
 &\implies \\
 S_n &= \frac{n}{2}(u_1 + u_n)
 \end{aligned}$$

Substitute u_n and you get the other printed form. Which one is shorter depends on what you were given: u_n known $\rightarrow$ the first, d known $\rightarrow$ the second.

The sum is not the term

$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$ is a **sum**. Its n th term is n . The whole of Part 1 rests on keeping those two apart, because the exam puts them in the same sentence: "*the n th triangular number is $\frac{n(n+1)}{2}$* " — a

triangular number is a sum, and the thing being summed is the sequence $1, 2, 3, \dots$

Task 1 ● — May 2022 TZ2 Paper 1
Q1, 5 marks

The n th term of an arithmetic sequence is given by $u_n = 15 - 3n$.

- (a)** State the value of the first term, u_1 .
(b) Given that the n th term of this sequence is -33 , find the value of n . **(c)** Find the common difference, d .

The check builds a progression out of your u_1 and your d together and asks whether it reproduces $15 - 3n$. Only one pair does, so (a) and (c) are pinned by the same line.

```

qla = ...           # the first term
qlb = ...           # the value of n for which
the term is -33
qlc = ...           # the common difference

mine = progression(qla, qlc)
# your first term, your step

verify_term('1a, 1c', 15 - 3 * n, mine, n)
# does it come out as 15 - 3n?
verify_term('1b', -33, mine, qlb)
# and is your n where it reaches -33?

```

Task 2 — November 2025 TZ1 Paper 1 Q1, 6 marks

The 1st and 5th terms of an arithmetic sequence are 36 and 12 respectively.

(a) Find the 13th term of this arithmetic sequence.

The sum of the first n terms of this arithmetic sequence is zero.

(b) Find the value of n .

The common difference is asked here too, although the paper does not: a wrong d and a wrong 13th term are two different mistakes, and it is worth knowing which one you made.

*In (b) there is a second answer the markscheme condones and the check does not: the sum of **no** terms is also zero. An index counts terms.*

```
q2d = ...           # the common difference
q2a = ...           # the 13th term
q2b = ...           # the number of terms whose
sum is zero

mine = progression(36, q2d)
# the first term is given

verify_term('2 (d)', 12, mine, 5)
# is the 5th term of yours 12?
verify_term('2a', q2a, mine, 13)
verify_total('2b', 0, mine, q2b)
```

Task 3 — May 2025 TZ1 Paper 2 Q10 (a)-(c), 6 marks

Rectangular playing cards are stacked in the shape of a pyramid with n rows. Some cards are placed horizontally and some are stacked at an angle of 60° to the horizontal. A pyramid with one row is two cards; a pyramid with two rows is seven; a pyramid with three rows adds another row below that.

Let t_n be the number of cards in a pyramid with n rows.

(a) Write down t_3 . **(b)** Find t_4 . **(c)** Show that $t_n = \frac{n(3n+1)}{2}$.

*Part (c) is a "show that", so the answer is printed in the question and there is nothing to hand over. What is asked instead is the step before it: **how many cards row n takes on its own**. That is the sequence being summed, and once you have it, (c) is one substitution.*

Row 1 is two cards. Each row below adds the same number more than the row above — that is what makes the total an arithmetic series.

```
q3r = ...          # the number of cards in row
n alone, in terms of n
q3a = ...          # t3
q3b = ...          # t4

rows = terms(lambda i: q3r.subs(n, i) if q3r
is not Ellipsis else ...)

verify_total('3c', n * (3 * n + 1) / 2, rows,
n)  # your rows add up to that
verify_total('3a', q3a, rows, 3)
verify_total('3b', q3b, rows, 4)
```

Part 2. Finding the progression

Theory: two facts are two equations

A progression is two numbers. Everything the paper tells you about it is a fact about those two numbers, and each fact is one equation.

what the paper says	the equation
the 7th term is 6	$u_1 + 6d = 6$
$S_8 = 8$	$\frac{8}{2}(2u_1 + 7d) = 8$
the 6th and 12th add to 24	$(u_1 + 5d) + (u_1 + 11d) = 24$
$u_8 = S_8$	$u_1 + 7d = \frac{8}{2}(2u_1 + 7d)$

Write both, solve, substitute back. That last step is not optional: it costs ten seconds and catches every sign slip you are going to make.

The trap is using one fact twice. " $u_8 = S_8 = 8$ " looks like one statement and is two: the eighth term is 8, **and** the first eight terms add to 8. Use only

one of them and the system has no unique solution.

Theory: from the sum back to the terms

Sometimes the paper gives S_n as a formula and asks for the terms. Two lines do the whole job:

$$u_1 = S_1, \quad u_n = S_n - S_{n-1}$$

The first because the sum of one term is that term. The second because S_n and S_{n-1} differ by exactly what the n th term added.

$$\begin{aligned} S_n &= 3n^2 - n \\ \implies u_n &= (3n^2 - n) - (3(n-1)^2 - (n-1)) \\ &= 6n - 4 \end{aligned}$$

u_n is **not** S_n/n . That is the average of the terms.

A quadratic S_n is the fingerprint of an arithmetic sequence, and the coefficients say which one: if $S_n = an^2 + bn$ then $d = 2a$, because the gap between consecutive

terms of a quadratic grows by twice its leading coefficient. You never have to remember that — $S_2 - 2S_1$ gives it in one line — but seeing it once explains why the sum comes out quadratic at all.

Task 4 — **May 2021 TZ1 Paper 1 Q2, 5 marks**

Consider an arithmetic sequence where $u_8 = S_8 = 8$. Find the value of the first term, u_1 , and the value of the common difference, d .

Both checks are backward: nothing in them is an answer except what sits inside `progression(...)`, and both eights came from the question.

```
q4u = ...           # the first term
q4d = ...           # the common difference

mine = progression(q4u, q4d)

verify_term('4 (u8)', 8, mine, 8)
# is the 8th term of yours 8?
verify_total('4 (S8)', 8, mine, 8)
# and do the first eight add to 8?
```

Task 5 — November 2025 TZ3

Paper 1 Q1, 5 marks

The 7th term of an arithmetic sequence is 6. The sum of the 6th term and the 12th term is 24.

Find the first term and the common difference.

The second condition is not a partial sum — it is two particular terms added — so the cell reads it out with `term(...)` and hands the check what is left over.

```
q5u = ...          # the first term
q5d = ...          # the common difference

mine = progression(q5u, q5d)

# the second condition names two terms, not a
sum: take one of them off
# the 24 and ask the check about the other
rest = ... if blank(q5u, q5d) else 24 -
term(mine, 12)

verify_term('5 (u7)', 6, mine, 7)
verify_term('5 (u6 + u12)', rest, mine, 6)
```

Task 6 — November 2023 TZ1

Paper 1 Q3, 7 marks

The sum of the first n terms of an arithmetic sequence is given by $S_n = pn^2 - qn$, where p and q are positive constants.

It is given that $S_4 = 40$ and $S_5 = 65$.

- (a)** Find the value of p and the value of q .
(b) Find the value of u_5 .

`sums(...)` turns your formula for the sums into the progression behind it: the first term is S_1 , and the step is $S_2 - 2S_1$. Both parts are then checked against the question's own 40 and 65.

```
q6p = ...          # p
q6q = ...          # q
q6b = ...          # the fifth term

mine = sums(lambda i: q6p * i ** 2 - q6q * i)

verify_total('6a (S4)', 40, mine, 4)
verify_total('6a (S5)', 65, mine, 5)
verify_term('6b', q6b, mine, 5)
```

Task 7 — **May 2023 TZ1 Paper 1**
Q10 (a)-(c), 9 marks

Consider the arithmetic sequence $u_1, u_2, u_3, \dots$. The sum of the first n terms of this sequence is given by $S_n = n^2 + 4n$.

(a) (i) Find the sum of the first five terms. **(ii)** Given that $S_6 = 60$, find u_6 . **(b)** Find u_1 . **(c)** Hence or otherwise, write an expression for u_n in terms of n .

Here S_n is given, so the progression behind it is not your answer — it is the question's. All four parts are checked forward against it, and (c) is checked at several values of n at once: a formula has to be right everywhere, not at one lucky index.

```
q7a = ...      # S5
q7b = ...      # u6
q7c = ...      # u1
q7d = ...      # u_n, in terms of n

given = sums(lambda i: i ** 2 + 4 * i)
# the sums the question gives

verify_total('7a(i)', q7a, given, 5)
verify_term('7a(ii)', q7b, given, 6)
verify_term('7b', q7c, given, 1)
verify_term('7c', q7d, given, n)
```

Part 3. The progression as a condition

Theory: one equation, read two ways

Up to here the sequence has been something you walk along. Now it becomes something you **impose**, and the whole of Part 3 is one equation:

$$u_2 - u_1 = u_3 - u_2$$

Read it forwards and it is a test: compute both sides and see whether they agree. Read it as an equation in a letter and it is a tool: it picks out the value of that letter for which the sequence *is* arithmetic.

The same statement rearranges into the form the markschemes prefer:

$$u_2 = \frac{u_1 + u_3}{2}$$

— **the middle term is the average of its neighbours.** Three quantities are in

arithmetic sequence exactly when the middle one is their mean.

Showing it holds for every n is a different job

"Show that $R_1, R_2, R_3, \dots$ form an arithmetic sequence" is not answered by checking three of them. The markscheme is explicit:

"Award M0 for consideration of special cases, for example R_3 and R_2 ."

What is wanted is $u_{n+1} - u_n$ in general, and then the sentence that earns the last mark: **and this does not depend on n , so the difference is constant.** Doing the algebra and not saying that costs an R1.

Theory: quantities that have no index

The last step of the topic drops the sequence entirely. Take $L(x) = mx + c$ and let r be its root. The paper calls L AS-

linear when m , r and c — **in that order** — are in arithmetic sequence.

There are no u_1 , no u_5 , no n anywhere.
There is one condition,

$$r - m = c - r,$$

and everything else is algebra: substitute the root of L in terms of m and c , clear the fraction, and a single equation in two letters comes out. That equation carries the rest of a thirty-one-mark investigation, and deriving it is the first thing the investigation asks for.

Order matters, and it is the first thing to check. m, r, c and m, c, r give different equations and different answers.

Task 8 — **May 2025 TZ3 Paper 1**
Q10 (a), 5 marks

Consider the sequence $\{u_n\}$ whose first three terms are

$$\begin{aligned}u_1 &= k - 5, \\u_2 &= 3 - 2k, \\u_3 &= 5k + 3, \\k &\in \mathbb{R}\end{aligned}$$

Consider the case when $\{u_n\}$ is arithmetic.

(a) (i) Find the value of k . **(ii)** Hence, or otherwise, find u_3 .

The first check has no answer of its own in it: your k goes into the three terms, and the question is whether they then have a constant difference. The second walks the progression those terms define and asks for its third member.

```
q8k = ...          # the value of k
q8u = ...          # the third term

def at(value):
    # the three terms of the question, at
    your k
    return [Ellipsis] * 3 if blank(value)
    else [value - 5,
          3 - 2 * value,
          5 * value + 3]

three = at(q8k)

verify_arithmetic('8a(i)', three)
verify_term('8a(ii)', q8u, terms(lambda i:
three[i - 1]), 3)
```

Task 9 — **May 2024 TZ2 Paper 1 Q10 (a),(d)(i), 4 marks, and May 2023 TZ2 Paper 1 Q12 (d), 3 marks**

Part one. Consider the arithmetic sequence $a, p, q, \dots$, where $a, p, q \neq 0$. It is given that $q = 1$ and $a = 9$.

(a) Find p , and write down the first four terms of the sequence.

(The paper asks first for the identity $2p - q = a$, which is the same condition rearranged: the middle term is the mean of its neighbours.)

Part two. The regions bounded by the curve $y = \cos \sqrt{x}$ and the x -axis are $R_1, R_2, R_3, \dots$, and their areas turn out to be $R_n = 4n\pi$.

(b) Show that the areas form an arithmetic sequence, and find the common difference.

verify_step refuses to answer the second question before the first: it checks that the difference is constant, and only then that it is yours.

```

q9p = ...          # the second term p, when a
= 9 and q = 1
q9t = [...]        # the first four terms
q9d = ...          # the common difference of
the areas R1, R2, R3, ...

verify_arithmetic('9a (p)', [9, q9p, 1])
# 9, your p, 1 – a constant step?
verify_start('9a (terms)', q9t, terms(lambda
i: [9, q9p, 1][i - 1]))
verify_step('9b', q9d, [4 * i * pi for i in
(1, 2, 3, 4)])

```

Task 10 — May 2023 TZ2 Paper 3 Q2 (a),(b)(ii),(c), 9 marks

Consider $L(x) = mx + c$ for $x \in \mathbb{R}$, where $m, c \in \mathbb{R}$ and $m, c \neq 0$. Let $r \in \mathbb{R}$ be the root of $L(x) = 0$, so that $r = -\frac{c}{m}$.

If m , r and c , **in that order**, are in arithmetic sequence, then L is said to be an **AS-linear function**.

(a) Show that $L(x) = 2x - 1$ is an AS-linear function, by giving the common difference of the sequence it makes. **(b)** Given that $L(x) = mx + c$ is AS-linear, show that

$$L(x) = mx - \frac{m^2}{m + 2}$$

by finding c in terms of m . **(c)** There are only three **integer** sets of values of m , r and c that form an AS-linear function. One of them is $L(x) = -x - 1$. Use part (b) to determine the other two.

Part (b) is checked without c ever being compared to anything: your expression goes into the triple $(m, -c/m, c)$, and the triple has to be arithmetic at every m the check tries. Part (c) is checked the same way, one function at a time, with the extra demand that all three numbers come out whole.

```

q10d = ...          # the common difference of
the sequence made by  $L(x) = 2x - 1$ 
q10c = ...          # c in terms of m
q10f = [...]        # the other two AS-linear
functions, as expressions in x

def triple(slope, constant):
    # the sequence the definition makes: the
    slope, the root, the constant
    return ([Ellipsis] * 3 if blank(slope,
constant)
            else [slope, -constant / slope,
constant])

verify_step('10a', q10d, triple(2, -1))
verify_arithmetic('10b', triple(m, q10c), m,
(1, 3, -5))

for got in q10f:
    verify_arithmetic('10c',
triple(got.coeff(x, 1), got.coeff(x, 0))
            if got is not Ellipsis
else [...])

```

Part 4. When n is the unknown

Theory: the sum stops growing

S_n is a quadratic in n , so if $d < 0$ it has a maximum. You can find it by completing the square, and you will get a fractional n — which is no answer at all, because n counts terms.

The honest route needs no quadratic:

The sum grows exactly while the terms are positive.

So solve $u_n \geq 0$, take the largest whole n , and substitute. If some term is exactly zero, two consecutive sums are equal and both indices are correct answers.

The same question on Paper 1 and on Paper 2

Task 2(b) — *the sum of the first n terms is zero* — is Paper 1, and it is one line of algebra: $\frac{n}{2}(36 + u_n) = 0$ needs $u_n = -36$, so $n = 13$.

Task 11 — *find the maximum value of S_n* — is Paper 2, and a table of S_n answers it in ten seconds. But the value must then be taken by substituting the whole n back into the formula, not read off the graph's vertex: the markscheme deducts the final mark for (13.1582..., 547.119...) by name.

Theory: when n must be a whole number and nothing else

Some questions cannot be turned into an equation at all:

- *the smallest number greater than 1 that is both triangular and pentagonal*
- *the fewest rows so that the cards come from whole packs of 52*

Both are conditions on integers, and algebra does not solve those. Every markscheme in the archive for this kind of question says the same thing: **uses a table of values**. Build the table, read it, stop.

The commonest slip is answering a nearby question. *"How many rows so that no cards are left over"* is not *"how many packs"*; the answers are 13 and 5, and only one of them is asked for.

Task 11 ● — **May 2021 TZ2 Paper 2**
Q2, 5 marks

An arithmetic sequence has first term 60 and common difference -2.5 .

(a) Given that the k th term of the sequence is zero, find the value of k .

Let S_n denote the sum of the first n terms of the sequence.

(b) Find the maximum value of S_n .

`verify_peak` finds the greatest sum by adding term after term and keeping the best. Because $u_{25} = 0$, two consecutive sums are equal here and both indices are right — the check accepts either.

```

q11k = ...          # the index of the zero term
q11s = ...          # the maximum value of Sn

mine = progression(60, -2.5)

verify_term('11a', 0, mine, q11k)
# is the term at your k zero?
verify_peak('11b', q11s, mine)

```

Task 12 — May 2022 TZ1 Paper 3 Q1 (d),(e), 8 marks, and May 2025 TZ1 Paper 2 Q10 (e), 2 marks

Part one. The n th pentagonal number can be represented by the arithmetic series

$$P_5(n) = 1 + 4 + 7 + \cdots + (3n - 2)$$

(a) Show that $P_5(n) = \frac{n(3n - 1)}{2}$, by giving the n th term of the series being added.

(b) By using a suitable table of values or otherwise, determine the smallest positive integer, greater than 1, that is both a triangular number and a pentagonal number, **and say which triangular number and which pentagonal number it is.** (The n th triangular number is $P_3(n) = \frac{n(n + 1)}{2}$.)

Part two. A complete pyramid stack of playing cards with n rows uses $t_n = \frac{n(3n+1)}{2}$ cards. A stack is built from full packs of 52 with no cards left over.

(c) Find the minimum number of rows in this stack, **and how many full packs it uses.**

Part (a) is a "show that", so what is asked is again the sequence being summed. Parts (b) and (c) are the two questions in the archive that algebra cannot answer, and both are checked backwards: your number is put back into the sequence, and the check asks whether it lands where you said. The two positions and the pack count are asked for that reason — without them there is nothing to check against but the answer itself.

*What the check cannot do is confirm that yours is the **smallest**. It will accept the 40th triangular number that is also pentagonal just as happily. That mark is yours; the table is what earns it.*

```

q12r = ...          # the nth term of the
pentagonal series, in terms of n
q12t = ...          # the smallest number > 1
that is both triangular and pentagonal
q12i = ...          # which triangular number
that is
q12j = ...          # which pentagonal number
that is
q12n = ...          # the minimum number of rows
q12k = ...          # how many full packs of 52
that stack uses

pent = terms(lambda i: q12r.subs(n, i) if
q12r is not Ellipsis else ...)
tri = terms(lambda i: i)
# 1 + 2 + 3 + ... is the triangular one
rows = terms(lambda i: 3 * i - 1)
# the card rows, from task 3
cards = ... if blank(q12k) else 52 * q12k

verify_total('12a', n * (3 * n - 1) / 2,
pent, n)
verify_total('12b (triangular)', q12t, tri,
q12i)
verify_total('12b (pentagonal)', q12t, pent,
q12j)
verify_total('12c', cards, rows, q12n)
# whole packs, and no cards over

```

Part 5. Terms that are logarithms

Theory: the log laws come first

Nothing about the sequence changes when its terms are logarithms. What changes is that you cannot compare them until they are written the same way.

$$9 + \ln 9, \quad 5 + \ln 3, \quad 1 + \ln 1$$

look like three unrelated things until $\ln 9 = 2 \ln 3$ and $\ln 1 = 0$ turn them into

$$9 + 2 \ln 3, \quad 5 + \ln 3, \quad 1 + 0 \cdot \ln 3$$

and now the difference is visible: $-4 - \ln 3$, the same both times.

Three laws do all the work in this topic:

$$\ln a^k = k \ln a, \quad \ln \frac{1}{a} = -\ln a, \quad \ln 1 = 0$$

The second one is where the marks go. $\ln \frac{1}{x^3}$ is $-3 \ln x$, not $3 \ln x$, and the minus sign is the whole answer.

Then the sum, and $\ln x$ cancels

When every term carries a factor of $\ln x$, so does the sum — and if the equation you are solving has $\ln x$ on both sides, it divides out and leaves a plain equation in n . That is the point of the whole construction: the logarithm is scaffolding, and the question underneath is a quadratic.

A last check that costs nothing: an arithmetic sequence has a constant **difference**. If you find yourself dividing consecutive terms, you have started answering a geometric question.

Task 13 — May 2024 TZ2 Paper 1 Q10 (e), 6 marks

The first three terms of an arithmetic sequence u_n are

$$u_1 = 9 + \ln 9,$$

$$u_2 = 5 + \ln 3,$$

$$u_3 = 1 + \ln 1$$

(a) Find the common difference of the sequence in terms of $\ln 3$. **(b)** Show that

$$\sum_{i=1}^{10} u_i = -90 - 25 \ln 3.$$

Part (b) is a "show that", and the printed answer is used as the question's own number: the check builds the progression out of your first term and your common difference, adds ten of them, and compares with $-90 - 25 \ln 3$. Write d wrongly and only that line fails.

```
q13d = ...          # the common difference, in
terms of ln 3
q13s = ...          # the sum of the first ten
terms

mine = progression(9 + ln(9), q13d)
# the first term is given

verify_step('13a', q13d, [9 + ln(9), 5 +
ln(3), 1 + ln(1)])
verify_total('13b', q13s, mine, 10)
verify_exact('13b (exact)', q13s, -90 - 25 *
ln(3))
```

Trainer: name the technique in five seconds

Twelve openings. Do not compute anything — say only **which move you**

would make first.

code	technique
term	u_1 and d are known; use $u_n = u_1 + (n - 1)d$
sum	something is being added up; use $S_n = \frac{n}{2}(u_1 + u_n)$
two	neither u_1 nor d is given; write two equations
back	S_n is given as a formula; use $u_n = S_n - S_{n-1}$
difference	impose or prove $u_2 - u_1 = u_3 - u_2$
coefficients	named quantities with no index put in sequence
peak	the greatest sum; the terms turn negative
integer	n has to be whole; build a table
log	the terms are logarithms; use the log laws first

1. An arithmetic sequence has $u_1 = 7$ and $d = 4$. Find the 20th term.
2. Find $1 + 2 + 3 + \cdots + 100$.
3. An arithmetic sequence has $u_4 = 11$ and $u_9 = 26$. Find u_1 and d .
4. The sum of the first n terms is $S_n = 3n^2 - 2n$. Find u_5 .

5. The sequence $k + 1, 2k, 7k - 6$ is arithmetic. Find k .
6. $L(x) = mx + c$ has root r , and m, r, c are in arithmetic sequence. Find c in terms of m .
7. An arithmetic sequence has $u_1 = 80$ and $d = -6.32$. Find the greatest value of S_n .
8. Find the smallest n for which $\frac{1}{2}n(3n + 1)$ is a multiple of 52.
9. The first three terms are $\ln x, \frac{2}{3} \ln x, \frac{1}{3} \ln x$. Find the common difference.
0. The 8th term of an arithmetic sequence equals the sum of its first eight terms, and both are 8. Find u_1 .
1. A pyramid of cards has 2 cards in the top row and 3 more in each row below. How many cards in six rows?
2. Show that the areas $R_n = 4n\pi$ form an arithmetic sequence.

```
answers = {
    1: '', 2: '', 3: '', 4: '', 5: '', 6: '',
    7: '', 8: '', 9: '', 10: '', 11: '', 12:
    '',
}

trigger_check(answers, {1: '599f7703547e', 2:
'09f5ffef2830', 3: '3fc4ccfe7458', 4:
'3c482346f375', 5: '69ea741e98fd', 6:
'7dbbfe42465d', 7: '9ee9e606474e', 8:
'2dba2454d800', 9: '836ff184e7b4', 10:
'3fc4ccfe7458', 11: '09f5ffef2830', 12:
'69ea741e98fd'})
```

On the clock — *May 2022 TZ1* *Paper 1 Q10 (b), 12 marks*

Twelve marks, eighteen minutes. No hints this time.

Consider the series

$$\ln x + p \ln x + \frac{1}{3} \ln x + \dots$$

where $x \in \mathbb{R}$, $x > 1$ and $p \in \mathbb{R}$, $p \neq 0$.

Consider the case where the series is **arithmetic** with common difference d .

(i) Show that $p = \frac{2}{3}$. **(ii)** Write down d in the form $k \ln x$, where $k \in \mathbb{Q}$. **(iii)** The sum

of the first n terms of the series is $\ln\left(\frac{1}{x^3}\right)$. Find the value of n .

Attempt log

date	time	result

```
qt_p = ...          # p
qt_d = ...          # the common difference, in
the form k*ln(x)
qt_n = ...          # the number of terms

mine = progression(ln(x), qt_d)
# the first term is given
three = [...] if blank(qt_p) else [ln(x),
qt_p * ln(x), ln(x) / 3]

verify_arithmetic('timer (i)', three)
verify_step('timer (ii)', qt_d, three)
verify_total('timer (iii)', ln(1 / x ** 3),
mine, qt_n)
```



Solutions

Work these only after you have your own answer, or you are reading, not practising.

1 (a) $u_1 = 15 - 3(1) = \boxed{12}$.

The trap is answering 15. That is u_0 , and the sequence starts at $n = 1$.

1 (b) $15 - 3n = -33 \implies 3n = 48 \implies \boxed{n = 16}$.

1 (c) Two consecutive terms differ by the coefficient of n :

$$u_2 - u_1 = (15 - 6) - (15 - 3) = \boxed{-3}$$

The markscheme accepts "*recognize gradient is -3* " on its own — the sequence is a linear function of n , and its gradient is d .

2 (a) From $u_1 = 36$ to $u_5 = 12$ there are **four** steps:

$$12 = 36 + 4d \implies d = \boxed{-6}$$

Then eight more steps to the 13th term:

$$u_{13} = u_5 + 8(-6) = 12 - 48 = \boxed{-36}$$

or, equally, $u_{13} = 36 + 12(-6) = -36$.

2 (b) $S_n = \frac{n}{2}(u_1 + u_n) = 0$ with $n \neq 0$ forces $u_1 + u_n = 0$, so $u_n = -36$ — and by (a) that is the 13th term:

$$\boxed{n = 13}$$

The symmetric way is prettier: $36 + 30 + \cdots + 0 + \cdots - 30 - 36$. Everything cancels in pairs around the zero term, and the zero term is the 7th, so there are $6 + 1 + 6 = 13$ terms.

The markscheme condones $n = 0, 13$. The check does not, and says why: an index counts terms.

3 (c) Row 1 takes 2 cards. Every row below takes 3 more than the row above — two more slanted cards and one more horizontal. So

$$\text{row } n = 3n - 1 \quad (2, 5, 8, 11, \dots)$$

and t_n is the sum of that arithmetic series:

$$t_n = \frac{n}{2}(2 + (3n - 1)) = \frac{n}{2}(3n + 1)$$

3 (a) $t_3 = 2 + 5 + 8 = \boxed{15}$.

3 (b) $t_4 = 15 + 11 = \boxed{26}$.

The mistake worth naming: adding 10 instead of 11. The fourth row has $3(4) - 1 = 11$ cards, not 10.

4 Two facts, two equations:

$$u_1 + 7d = 8, \quad \frac{8}{2}(2u_1 + 7d) = 8$$

The second gives $2u_1 + 7d = 2$.
Subtracting the first,

$$\begin{aligned} u_1 &= 2 - 8 \\ &= -6, \quad 7d \\ &= 8 - (-6) \\ &= 14 \\ &\implies d \\ &= 2 \end{aligned}$$

$$\boxed{u_1 = -6, \quad d = 2}$$

Faster still: $S_8 = \frac{8}{2}(u_1 + u_8) = 4(u_1 + 8) = 8$ gives $u_1 = -6$ in one line, because u_8 is already known to be 8.

5 $u_7 = 6$ gives $u_1 + 6d = 6$. For the second condition, either expand both terms,

$$\begin{aligned}
 (u_1 + 5d) + (u_1 + 11d) &= 24 \\
 &\implies 2u_1 + 16d \\
 &= 24 \\
 &\implies u_1 + 8d \\
 &= 12,
 \end{aligned}$$

or notice that u_6 and u_{12} sit one step below and five steps above u_7 :

$$\begin{aligned}
 u_6 &= 6 - d, & u_{12} \\
 &= 6 + 5d \\
 &\implies 12 + 4d \\
 &= 24 \\
 &\implies d \\
 &= 3
 \end{aligned}$$

Then $u_1 = 6 - 6(3) = \boxed{-12}, \boxed{d = 3}$.

The second route is the markscheme's METHOD 2 and it is three lines shorter.

Measuring from a term you already know, rather than from u_1 , is worth having as a habit.

6 (a) $S_4 = 16p - 4q = 40$ and $S_5 = 25p - 5q = 65$, that is

$$4p - q = 10, \quad 5p - q = 13$$

Subtracting, $p = 3$; then $q = 4(3) - 10 = 2$.

$$\boxed{p = 3, \quad q = 2}$$

6 (b) $u_5 = S_5 - S_4 = 65 - 40 = \boxed{25}$.

Nothing about p and q is needed for (b) at all. It is the fastest mark on the paper and the one most often done the long way.

7 (a)(i) $S_5 = 25 + 20 = \boxed{45}$.

(ii) $u_6 = S_6 - S_5 = 60 - 45 = \boxed{15}$.

7 (b) $u_1 = S_1 = 1 + 4 = \boxed{5}$.

7 (c) $u_n = S_n - S_{n-1}$:

$$\begin{aligned}
& (n^2 + 4n) \\
& \quad - ((n-1)^2 + 4(n-1)) = n^2 \\
& \quad + 4n \\
& \quad - n^2 \\
& \quad + 2n \\
& \quad - 1 \\
& \quad - 4n \\
& \quad + 4 \\
& = \boxed{2n + 3}
\end{aligned}$$

Or through d : $u_2 = S_2 - S_1 = (4 + 8) - 5 = 7$, so $d = 7 - 5 = 2$ and $u_n = 5 + 2(n-1) = 2n + 3$. The check accepts both forms, and $n + (n + 3)$ as well — it walks the sequence, it does not read the expression.

8 (a)(i) Arithmetic means the two gaps are equal:

$$(3 - 2k) - (k - 5) = (5k + 3) - (3 - 2k)$$

$$8 - 3k = 7k \implies 10k = 8 \implies k = \boxed{\frac{4}{5}}$$

The mean form is quicker:
 $\frac{(k-5) + (5k+3)}{2} = 3 - 2k$ gives $3k - 1 = 3 - 2k$ in one step.

(ii) $u_3 = 5 \left(\frac{4}{5}\right) + 3 = \boxed{7}$.

The terms are $-\frac{21}{5}, \frac{7}{5}, 7$ — a common difference of $\frac{28}{5}$. Substituting k into the wrong expression is the standard slip here; u_3 is $5k + 3$.

9 (a) In a, p, q the middle term is the mean:

$$p = \frac{a + q}{2} = \frac{9 + 1}{2} = \boxed{5}$$

which is the identity $2p - q = a$ rearranged. The common difference is -4 , so the first four terms are

$$\boxed{9, 5, 1, -3}$$

9 (b) $R_{n+1} - R_n = 4(n+1)\pi - 4n\pi = \boxed{4\pi}$.

This does not depend on n , **so the difference is constant and the areas form an arithmetic sequence** — that sentence is the R1, and the markscheme awards M0 for checking R_3 and R_2 instead.

10 (a) $L(x) = 2x - 1$ has $m = 2$, $c = -1$ and root $r = \frac{1}{2}$. The sequence 2, $\frac{1}{2}$, -1 has

$$d = \frac{1}{2} - 2 = -1 - \frac{1}{2} = \boxed{-\frac{3}{2}}$$

Equal both ways, so it is arithmetic and L is AS-linear.

10 (b) With $r = -\frac{c}{m}$, the condition $r - m = c - r$ becomes

$$\begin{aligned} -\frac{c}{m} - m &= c + \frac{c}{m} \\ -2\frac{c}{m} &= c + m \\ &\implies -2c \\ &= cm + m^2 \\ &\implies m^2 + cm + 2c \\ &= 0 \end{aligned}$$

Solve for c :

$$c(m + 2) = -m^2 \implies c = \boxed{-\frac{m^2}{m + 2}}$$

and so $L(x) = mx - \frac{m^2}{m + 2}$. The restriction that comes with it is $m \neq -2$, and it is worth its own mark.

10 (c) c is an integer exactly when $m + 2$ divides m^2 . Since $m^2 = (m - 2)(m + 2) + 4$, that means $m + 2$ divides 4:

$$\begin{aligned}m + 2 &\in \{\pm 1, \pm 2, \pm 4\} \\ \implies m &\in \{-1, -3, 0, -4, 2, -6\}\end{aligned}$$

Discard $m = 0$, and check which give integer $r = -c/m$ as well. The three that survive are $m = -1$ (the one given, $L(x) = -x - 1$) and

$L(x) = -4x + 8,$	$L(x) = -3x + 9$
-------------------	------------------

with sequences $-4, 2, 8$ and $-3, 3, 9$ — differences 6 and 6.

11 (a) $u_k = 60 - 2.5(k - 1) = 0$ gives $2.5(k - 1) = 60$, so $k - 1 = 24$ and

$k = 25$

11 (b) The terms are positive up to the 24th, zero at the 25th and negative after, so the sum is greatest at $n = 24$ **and** at $n = 25$ — both give the same total:

$$S_{25} = \frac{25}{2}(60 + 0) = \boxed{750}$$

A table of S_n on the GDC gets there faster, but read the *value* off the table, not off the graph: the vertex of the parabola sits at a fractional n , and that answer is explicitly not accepted.

12 (a) The series $1 + 4 + 7 + \dots$ has $u_1 = 1$ and $d = 3$, so its n th term is

$$\boxed{3n - 2}$$

and the sum is

$$P_5(n) = \frac{n}{2}(1 + (3n - 2)) = \frac{n(3n - 1)}{2}$$

12 (b) Two tables side by side:

n	$P_3(n)$		m	$P_5(m)$
18	171		10	145
19	190		11	176
20	210		12	210
21	231		13	247

$$\boxed{210}$$

the 20th triangular number and the 12th pentagonal number. Stopping the table

too early is the whole difficulty: nothing smaller than 210 works, and 1 is excluded by the question.

12 (c) $t_n = \frac{1}{2}n(3n + 1)$ has to be a multiple of 52:

2,
7,
15,
26,
40,
57,
77,
100,
126,
155,
187,
222,
260

and $260 = 5 \times 52$ at $n = 13$, so the answer is 13 rows.

Two ways to lose this mark: solving $t_n = 52$ (which the markscheme rejects by

name), and answering 5 — the number of packs, not the number of rows.

13 (a) Rewrite every term over $\ln 3$:

$$u_1 = 9 + 2 \ln 3,$$

$$u_2 = 5 + \ln 3,$$

$$u_3 = 1 + 0$$

$$\begin{aligned}d &= u_2 - u_1 \\&= (5 - 9) + (1 - 2) \ln 3 \\&= \boxed{-4 - \ln 3}\end{aligned}$$

and $u_3 - u_2 = (1 - 5) + (0 - 1) \ln 3$ gives the same, as it must.

13 (b)

$$\begin{aligned}S_{10} &= \frac{10}{2} (2u_1 + 9d) \\&= 5 \left(2(9 + 2 \ln 3) + 9(-4 - \ln 3) \right) \\&= 5(18 + 4 \ln 3 - 36 - 9 \ln 3) = 5(-18 - 5 \ln 3) \\&= \boxed{-90 - 25 \ln 3}\end{aligned}$$

Both halves of the answer are needed. Cancelling the logarithms early —

because $\ln 1 = 0$ — loses the $-25 \ln 3$ entirely.

Timer

(i) Arithmetic means the two gaps are equal:

$$p \ln x - \ln x = \frac{1}{3} \ln x - p \ln x$$

Since $x > 1$, $\ln x \neq 0$ and it divides out:

$$p - 1 = \frac{1}{3} - p \implies 2p = \frac{4}{3} \implies p = \boxed{\frac{2}{3}}$$

(ii) $d = p \ln x - \ln x = \left(\frac{2}{3} - 1\right) \ln x = \boxed{-\frac{1}{3} \ln x}.$

(iii) First the right-hand side, and this is where the marks are:

$$\ln\left(\frac{1}{x^3}\right) = \ln x^{-3} = -3 \ln x$$

Now the sum:

$$S_n = \frac{n}{2} \left(2 \ln x + (n-1) \left(-\frac{1}{3} \ln x \right) \right) = -3 \ln x$$

Divide by $\ln x$ — it is on both sides, and this is the point of the whole construction:

$$\begin{aligned}
\frac{n}{2} \left(2 - \frac{n-1}{3} \right) &= -3 \\
&\implies \frac{n}{2} \cdot \frac{7-n}{3} \\
&= -3 \\
&\implies n(7-n) \\
&= -18
\end{aligned}$$

$$\begin{aligned}
n^2 - 7n - 18 &= 0 \\
&\implies (n-9)(n+2) \\
&= 0 \\
&\implies \boxed{n=9}
\end{aligned}$$

$n = -2$ is discarded: an index counts terms.

The markscheme's second method is worth seeing, because it needs no algebra at all. The terms are

$$\begin{aligned}
&\ln x, \\
&\frac{2}{3} \ln x, \\
&\frac{1}{3} \ln x, \\
&0, \\
&-\frac{1}{3} \ln x, \\
&-\frac{2}{3} \ln x, \\
&-\ln x, \\
&-\frac{4}{3} \ln x, \\
&-\frac{5}{3} \ln x, \\
&\dots
\end{aligned}$$

The first seven cancel in pairs around the zero and sum to nothing. The 8th and 9th add to $-3 \ln x$, which is exactly what was wanted. So $n = 9$ — and that symmetry is the same one that answered task 2(b).